

Session 2018-2019

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A. What are the different components of IoT? How does the Internet of Things (IoT) affect our everyday lives?

Components of IoT:

- Devices/Sensors: Gadgets that gather data, like smartwatches and security cameras.
- Connectivity: How these gadgets connect to the internet, using Wi-Fi, Bluetooth, or cellular networks.
- Data Processing: The data is analyzed either on the device, nearby computers, or in the cloud.
- User Interface: How we interact with IoT devices, such as through apps, websites, or voice commands.

How IoT Affects Our Everyday Lives:

- Smart Homes: Control lights, thermostats, and security systems remotely, making life easier and homes more secure.
- Health and Fitness: Wearable devices track activities, heart rate, and sleep to help you stay healthy.
- Transportation: IoT helps with real-time navigation, vehicle monitoring, and even self-driving cars.
- Workplaces: Automate tasks like lighting and heating, making offices more efficient and productive.
- Retail: Improve shopping with smart shelves, personalized ads, and stores without cashiers.
- Agriculture: Farmers use IoT to monitor crops and soil, improving resource use and crop growth.
- Cities: Manage traffic, air quality, and energy use to make city living better.

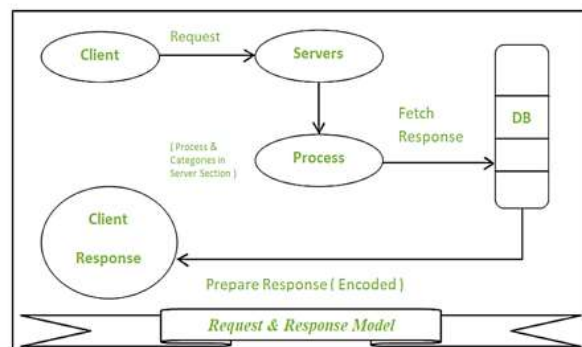
A. What is communication model in IoT? Explain different types of communication model in IoT with diagram and example.

The communication model in IoT explains how IoT devices talk to each other and share information. It includes the different ways and paths that devices use to connect and exchange data.

Different types of communication model:

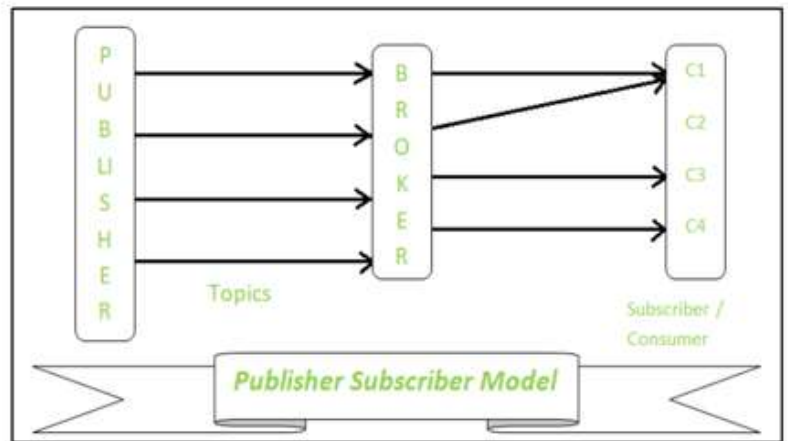
1. Request & Response Model

Example: Sending a spreadsheet to the printer



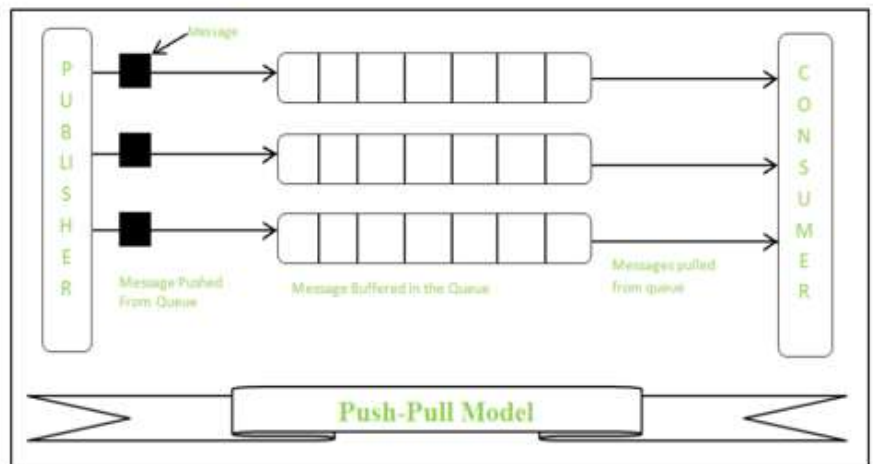
2. Publisher-Subscriber Model

Example: Public sensors with a massive base of uniform users that will use the data.



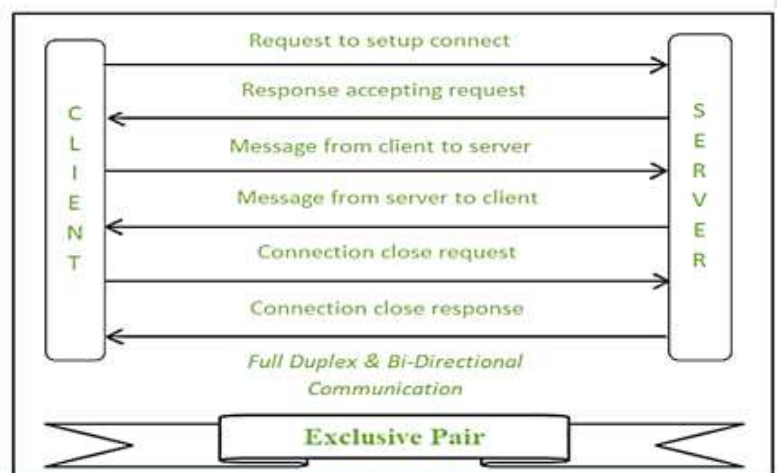
3. Push-Pull Model

Example: Queues help in decoupling the messaging between the producers and consumers.



4. Exclusive Pair

Example: The WebSocket-based communication API.



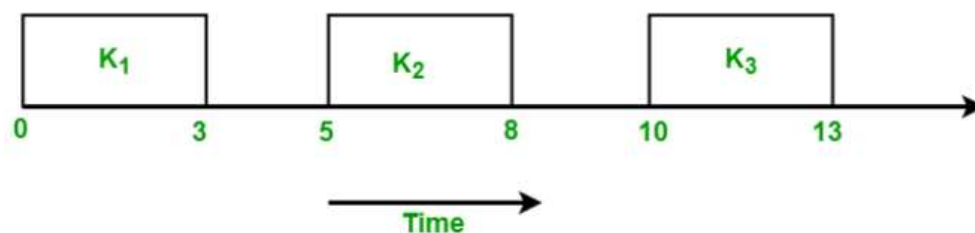
[C.] Write down the difference between Embedded and Real Time Systems.

Real-Time System	Non-Real-Time System
A real-time system is the one whose logical correctness is based on both the correctness of the outputs and their timeliness.	A non-real-time system is the one in which tasks are not associated with any time bounds or deadlines.
If a real-time system fails to meet the deadline, it may result in catastrophe or severe consequences.	There is no severe consequences if a system fails to meet a deadline.
Real-Time systems are deterministic.	It is non-deterministic. The arrival of event can't be determined.
It is time sensitive.	It is time insensitive.
It can't use virtual memory.	It can use concept of virtual memory.
It is dedicated to single work.	It is used in multi-user environment.
It has flat memory model.	It has protected memory mode.
It has low interrupt latency.	It has high interrupt latency.
It can be expressed as quantitative expression of time.	It can't be expressed as function of time.
It can be used in critical systems.	It can't be used in critical system. They are only used for general works.
Example: Satellite tracking, video conferencing etc.	Example: Batch processing jobs, old email services etc.

2

[A.] What is the periodic and aperiodic task model in real-time system? Explain with example.

Consider the task T_i with period = 5 and execution time = 3. Phase is not given so, assume the release time of the first job as zero. So, the job of this task is first released at $t = 0$ then it executes for 3s and then the next job is released at $t = 5$ which executes for 3s and then the next job is released at $t = 10$. So, jobs are released at $t = 5k$ where $k = 0, 1, \dots, n$.



Aperiodic real-time tasks occur at random intervals, which may even be zero between two tasks. These tasks are also referred to as sporadic, asynchronous, or reactive. They create dynamic situations that require flexible management.

In systems with resource constraints, handling tasks with bounded arrival time intervals is more feasible, while those with unbounded intervals present significant challenges.

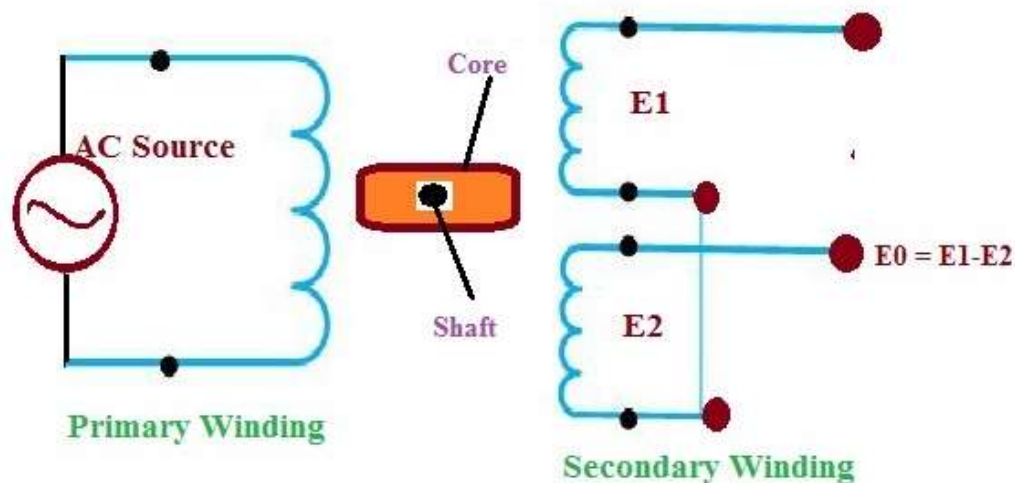
A practical example of an aperiodic real-time task is **typing on a keyboard**, where each keypress can occur at unpredictable times.

[B.] What is a RVDT? How does RVDT work step by step? Explain with RVDT construction diagram.

The RVDT stands for **Rotary Variable Differential Transformer**. It is one kind of electromechanical transducer used to give the linear o/p which is proportional to the i/p angular displacement.

RVDT Construction

- **Components:** RVDT has **a primary winding and two secondary windings on either side.**
- **Core:** A **soft iron core is attached to a shaft** and can rotate between these windings.



Working Conditions:

Null Position:

- ✓ **Position:** The core is centered.
- ✓ **Induced Voltages:** The voltages in the secondary windings (**E1** and **E2**) are equal but opposite.
- ✓ **Output Voltage:** The output (**E0**) is zero: $E0 = E1 - E2 = 0$

Clockwise Rotation:

- ✓ Position: The core rotates clockwise.
- ✓ Induced Voltages: E_1 becomes greater than E_2 .
- ✓ Output Voltage: The output is positive: $E_0 = E_1 - E_2 > 0$

Anticlockwise Rotation:

- ✓ Position: The core rotates anticlockwise.
- ✓ Induced Voltages: E_2 becomes greater than E_1 .
- ✓ Output Voltage: The output is negative: $E_0 = E_1 - E_2 < 0$

[C.] How many principles are there in Norman design principles? Write down Norman's design principles for IoT.

Norman's design principles consist of 6 principles.

Here's a mnemonic to help you remember Norman's design principles for IoT: "Very Fast Cats Make Clean Animals"

1. **Visibility:** Make important elements visible to users. For IoT, ensure that device status, controls, and data are easy to see and understand (e.g., status lights or app notifications).
2. **Feedback:** Provide immediate feedback for user actions. IoT devices should confirm actions like turning on/off or connecting to a network through lights, sounds, or app responses.
3. **Constraints:** Limit user interactions to prevent errors. For IoT devices, only allow valid inputs or commands, and use physical constraints like shape or size to guide users.
4. **Mapping:** Create a clear relationship between controls and their effects. For IoT, make sure that buttons or gestures map logically to their functions (e.g., swiping up to increase temperature).
5. **Consistency:** Maintain uniformity in design and interactions. All similar IoT devices should behave the same way, making it easier for users to switch between devices.
6. **Affordance:** Design elements should suggest their use. For IoT, controls like buttons or sliders should look like they can be pressed or moved, making it clear what actions are possible.

[D.] Briefly explain the most communication protocols used by IoT.

1. MQTT (Message Queuing Telemetry Transport): Sends small messages; great for low-power devices like sensors.
2. CoAP (Constrained Application Protocol): Works well with simple devices that have limited power and memory.
3. HTTP/HTTPS (Hypertext Transfer Protocol/Secure): Used to connect to websites but not ideal for devices with limited power.
4. Bluetooth/BLE (Bluetooth Low Energy): Short-range, low-power; used in fitness trackers and smartwatches.
5. Zigbee: Low-power, good for smart homes like controlling lights.

6. LoRaWAN (Long Range Wide Area Network): Long-distance communication; used in smart cities and agriculture.
7. Wi-Fi (Wireless Fidelity): High-speed, for devices like smart cameras but uses more power.
8. Z-Wave: Similar to Zigbee, often used in home automation.

3

[A.] Explain the characteristics of RTS.

➤ Time Constraints:

- time interval allotted for the response of the ongoing program.
- task should be completed within time interval.

➤ Correctness:

- produce correct result within the given time interval.
- If the result is not obtained within the given time interval then also result is not considered correct.

➤ Safety:

- perform for a long time without failures.
- recovers very soon
- failure occurs in the system and it does not cause any harm to the data and information.

➤ Concurrency:

- respond to a several number of processes at a time.
- responds accordingly to every task in short intervals.

➤ Distributed:

- all the components of the systems are connected in a distributed way.
- The real-time systems are connected in such a way that different components are at different geographical locations.
- operations are operated in distributed ways.

➤ Stability:

- Even when the load is very heavy, real-time systems respond in the time constraint
- real-time systems does not delay the result of tasks even when there are several task going on a same time.

[B.] Write down the difference between Sensors and Transducer.

Feature	Sensor	Transducer
Definition	A device which provides a usable output in response to a specified measurand.	A transducer is a device that converts <i>one form of energy into another</i> .
Function	Primarily used for <i>sensing and measuring physical parameters</i> (e.g., temperature, light, pressure).	Used for <i>energy conversion between different forms</i> (e.g., electrical to mechanical or vice versa).
Examples	Temperature sensor, pressure sensor, light sensor.	Microphone, loudspeaker, piezoelectric transducer.
Relation	All sensors are transducers.	Not all transducers are sensors (e.g., loudspeaker is a transducer but not a sensor).
Output	Usually an electrical signal (e.g., voltage, current).	Can be any form of energy (e.g., electrical, mechanical).

[C.] what is human computer interaction? Briefly explain most important components of HCI.

Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.

**1. The user**

- The user component refers to an individual or a group of individuals that participate in a common task. HCI studies users' needs, goals, and interaction patterns. It analyzes various parameters such as users' cognitive capabilities, emotions, and experiences to provide them with a seamless experience while interacting with computing systems.

2. The goal-oriented task

- A user operates a computer system with an objective or goal in mind. The computer provides a digital representation of objects to accomplish this goal.

3. The interface

- The interface is a crucial HCI component that can enhance the overall user interaction experience.
- Various interface-related aspects must be considered, such as interaction type (touch, click, gesture, or voice), screen resolution, display size, or even color contrast. Users can adjust these depending on the user's needs and requirements.

4. The context

- HCI is not only about providing better communication between users and computers but also about factoring in the context and environment in which the system is accessed.
- For example, while designing a smartphone app, designers need to evaluate how the app will visually appear in different lighting conditions (during day or night) or how it will perform when there is a poor network connection. Such aspects can have a significant impact on the end-user experience.
- Thus, HCI is a result of continuous testing and refinement of interface designs that can affect the context of use for the users.

[D.] What are the types of IoT security? How to secure IoT Devices?

Types of IoT Security

- ✓ **Network Security:** Protects data while it travels between devices.
- ✓ **Device Security:** Secures the physical device and its software.
- ✓ **Cloud Security:** Protects data stored in the cloud.
- ✓ **Application Security:** Keeps apps running on devices safe.
- ✓ **Data Security:** Encrypts data to prevent unauthorized access.
- ✓ **Identity and Access Management (IAM):** Manages who can access devices.

How to Secure IoT Devices

- ✓ **Change Default Passwords:** Use strong, unique passwords.
- ✓ **Update Firmware:** Regularly update software for security patches.
- ✓ **Use Separate Networks:** Isolate IoT devices from the main network.
- ✓ **Enable Device Authentication:** Use multi-factor authentication.
- ✓ **Disable Unused Features:** Turn off features you don't need.
- ✓ **Encrypt Data:** Secure data both in storage and during transmission.
- ✓ **Use Firewalls:** Block suspicious activities with firewalls.
- ✓ **Monitor Activity:** Keep an eye on logs for unusual behavior.